Monte Carlo simulation of stochastically heated grains and comparison with the Guhathakurta–Draine method

Abstract

We implement the Monte Carlo (MC) algorithm of Draine & Anderson (1985) for tracking the temperature of a single interstellar dust grain through individual photon-absorption events, extend it with a selectable heat-capacity backend, and reproduce Figures 24.5 and 24.6 of Draine (2011). The standalone implementation is kept for visualization; a subroutine-form API is then used to integrate $P(T)$ over a grain size distribution and build an emergent SED. The SED builder exposes a runtime flag `use_mc_for_large_grain`. In the default mode (`=F`), small (stochastic) grains use the Monte Carlo engine while large (equilibrium) grains use the analytic `calc_Teq` solver – the same hybrid logic the production pipeline uses, but with MC replacing the Guhathakurta & Draine (1989) matrix solver in the stochastic branch. In this mode the MC-based SED reproduces the production `sed_solve` output to within 1–3% in every band, once three corrections are applied to the SED builder: (i) a fractional-temperature cap on the linearized cooling step, which keeps the super-linear T^4 cooling from over-populating the hot tail of $P(T)$ (otherwise a $\sim 20\%$ mid-IR over-prediction for the tiniest PAHs); (ii) a single-grain energy-conservation rescale that forces each grain's grey bolometric emission to equal its absorbed power, removing a Planck-midpoint convexity bias; and (iii) charge-resolved PAH emission (neutral and cation summed separately, as the production solver does), which fixes a $\sim 23\%$ near-IR deficit. Setting the flag to `T` switches large grains to MC as well. The all-MC mode additionally needs an adaptive cutoff $\lambda_c(a) = hc/(\delta T_{\rm thresh} T_{\rm eq} C_{\rm tot}(T_{\rm eq}))$ (§7.1) in place of the static $\lambda_c = 1000\,\mu$m: without it the single-photon temperature jumps for large grains are too small to keep the trajectory near $T_{\rm eq}$ and the FIR is under-predicted by $\sim 30\%$. With all corrections the all-MC SED is indistinguishable from the default mode (both within 1–3% of the reference) and the single-grain $T_{\rm rad}^{\rm MC}/T_{\rm eq}$ ratio stays within $\pm 3\%$ across $a \in [0.03, 5]\,\mu$m. A second runtime flag, `mc_engine`, selects between the original fixed-grid histogram (log $[2, 5000]$ K, $N_{\rm hist} = 600$) and two adaptive-grid engines introduced in Section 7.: a deterministic two-pass engine that probes $T_{\rm min}/T_{\rm max}$ in a first pass and replays the trajectory with the same RNG state on a fitted grid, and a single- pass engine that buffers sub-step samples and bins them after the run. Both adaptive engines auto-select log versus linear spacing based on $T_{\rm max}/T_{\rm min}$, which gives a $\sim 50\times$ sharper resolution of the narrow equilibrium $P(T)$ for large grains where the fixed grid collapses to one populated bin. *All Monte Carlo results in this report (Figures 1–6) use the adaptive two-pass engine*; the fixed-grid engine is retained in the code for backward compatibility and is shown only as a baseline in Figure 6.

Note on the Mathis ISRF. The production `radfield.f90 :: J_Mathis`, which mc_pT shares with `sed_solve`, was switched on 2026-05-17 to the Draine's corrected 4000 K dilution factor (1.65×10^{-13}, Draine textbook 2011 Table 12.1; the literal Mathis 1983 10^{-13} value used previously is retained as `use_mathis_corrected = .false.` for reproducibility, with full discussion in `SEDust/docs/astrodust_sed_report.tex` §7.1.1, "Mathis 4000 K dilution correction"). At the fiducial $a = 0.1\,\mu$m astrodust grain the correction shifts $T_{\rm eq}$ by +0.22 K (+1.28%) and the FIR-peak emission per grain by +8.4%. The PIIM Figs. 24.5/24.6 reproductions (Figures 1, 2), the SED comparison against `sed_solve` (Figure 3), and the single-grain $P(T)$ comparisons (Figures 4, 5, 6) have all been regenerated with the new default and the adaptive two-pass MC engine; the 1–3% MC-vs-reference agreement is invariant to the choice of dilution factor because both sides shift together.

Contents

1. Introduction

A dust grain in the interstellar medium absorbs starlight photons discretely. When the grain is macroscopic ($a \gtrsim 0.03\,\mu$m), many absorption events occur per radiative cooling time and the grain sits at a well-defined equilibrium temperature $T_{\rm eq}(a)$ at which the mean absorbed power balances the radiative cooling. Smaller grains, however, do not have enough heat capacity to absorb the energy of a single starlight photon without making a large, transient excursion in temperature. Each photon absorption raises the grain's temperature by tens to hundreds of Kelvin; between events the grain cools radiatively back toward $T_{\rm eq}$. The temperature history $T(t)$ is a sequence of sharp upward spikes followed by long, slow exponential cooling tails, as illustrated in Figure 24.5 of Draine (2011).

Because the emission spectrum scales steeply with temperature ($\propto T^4$ in total luminosity, with strong wavelength dependence through the Planck function and the cross section), the spectral energy distribution (SED) of small grains is dominated by the high-temperature spikes, not by $T_{\rm eq}$. Quantitative treatment requires the probability distribution $P(T)$, which is the fraction of time the grain spends at each temperature in steady state.

There are two standard ways to obtain $P(T)$:

1. **Monte Carlo** (Draine & Anderson, 1985). Simulate the stochastic sequence of absorption events directly and accumulate a time-weighted histogram of $T(t)$.

2. **Discretized steady-state matrix solve** (Guhathakurta & Draine, 1989; Draine & Li, 2001). Discretize T into bins, build a transition matrix from absorption (upward) and Planck-averaged emission (downward) rates, and solve for the steady state.

The matrix method is more efficient and is what the production astrodust pipeline uses (`p_sub.f90::calc_P`). The MC method is conceptually simpler, has fewer numerical knobs, and provides a useful independent cross-check. This document records the MC implementation kept under `SEDust/mc/` and its validation.

Document layout. Section 2. derives the MC algorithm and the cooling ODE used between events. Section 3. describes the Fortran module structure and the selectable heat-capacity backend. Section 4. presents the reproduction of Draine (2011) Figures 24.5 and 24.6 and the energy-balance test. Section 5. describes the subroutine-form API and the SED builder that integrates $P(T)$ over a grain size distribution. Section 6. compares the MC-based SED against the Guhathakurta–Draine matrix solver on identical inputs.

2. Algorithm

2.1 Two-regime treatment of the absorbed radiation field

Consider a spherical grain of radius a in an isotropic radiation field of mean intensity $J_\lambda(\lambda)$. The radiative energy density per unit wavelength is

$$u_\lambda = \frac{4\pi}{c} J_\lambda \quad [\mathrm{erg\,cm^{-3}\,\mu m^{-1}}]. \tag{1}$$

The grain absorbs photons of wavelength λ at a rate per unit wavelength

$$\frac{dN_{\rm abs}}{dt\,d\lambda} = \pi a^2 Q_{\rm abs}(a,\lambda)\,\frac{c\,u_\lambda\,\lambda}{hc} = \frac{\pi a^2}{h}\,Q_{\rm abs}(a,\lambda)\,u_\lambda\,\lambda, \tag{2}$$

where the photon energy is $E = hc/\lambda$ and we have used $u_\lambda/E = (u_\lambda\,\lambda)/(hc)$ as the photon number density per unit wavelength, multiplied by c to get a photon flux density and by $\pi a^2 Q_{\rm abs}$ to get the absorption rate.

Draine & Anderson (1985) split this rate into a continuous part (long wavelengths, where photons arrive so frequently that the grain heat content increases smoothly) and a stochastic part (short wavelengths, where individual events are resolved). Pick a cutoff λ_c (default 1000 μm). The continuous-heating rate per unit surface area is

$$H(a) \equiv \frac{c}{4}\int_{\lambda_c}^{\infty} Q_{\rm abs}(a,\lambda)\, u_\lambda \, d\lambda \quad [\mathrm{erg\,cm^{-2}\,s^{-1}}]. \tag{3}$$

The stochastic event rate is

$$\Gamma_{\rm evt}(a) \equiv \int_0^{\lambda_c} \frac{dN_{\rm abs}}{dt\, d\lambda}\, d\lambda \quad [\mathrm{s^{-1}}]. \tag{4}$$

For the Mathis ISRF and a small grain, $\Gamma_{\rm evt}$ is dominated by ultraviolet photons; $H(a)$ is essentially the CMB contribution and is numerically negligible. Both contributions are nevertheless kept so that the same code handles dense-cloud and starlight-heated conditions uniformly.

The tabulated wavelength grid generally has one interval straddling λ_c. That interval is split at λ_c by linear interpolation of the integrand: its blueward part is added to $\Gamma_{\rm evt}$ (and to the photon-sampling CDF) and its redward part to $H(a)$, so the cutoff bin is dropped from neither term. The sampling CDF is built from the same split integral as $\Gamma_{\rm evt}$ and has its last node pinned at λ_c, so a sampled photon always satisfies $\lambda \le \lambda_c$. A grain with no absorption blueward of λ_c has $\Gamma_{\rm evt} = 0$ and belongs on the equilibrium path; `grain_setup_from_cabs` stops with a diagnostic rather than divide by a zero event rate.

2.2 Cooling ODE between events

Let $E_{\rm grain}(T)$ be the total grain enthalpy and $C_{\rm tot}(T) \equiv dE_{\rm grain}/dT$ its total heat capacity. Between absorption events, the grain's energy evolves according to

$$\frac{dE_{\rm grain}}{dt} = P_{\rm heat} - P_{\rm emit} = 4\pi a^2 \left[H(a) - \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4 \right], \tag{5}$$

where $\langle Q_{\rm abs}\rangle_T$ is the Planck-averaged absorption efficiency,

$$\langle Q_{\rm abs}\rangle_T \equiv \frac{\int Q_{\rm abs}(a,\lambda) B_\lambda(T)\, d\lambda}{\int B_\lambda(T)\, d\lambda}. \tag{6}$$

Dividing through by $C_{\rm tot}$ gives the cooling ODE used in the MC:

$$\boxed{\frac{dT}{dt} = \frac{4\pi a^2}{C_{\rm tot}(T)} \left[H(a) - \langle Q_{\rm abs}\rangle_T\, \sigma_{\rm SB}\, T^4 \right].} \tag{7}$$

Equation (7) is identical to Eq. (2.9) of Draine & Anderson (1985), which writes it in the equivalent form $dT/dt = (3/(aC_{\rm vol}(T)))\,(H - \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4)$ using the heat capacity per unit volume $C_{\rm vol} = C_{\rm tot}/(4\pi a^3/3)$. We use the total form because the enthalpy routine (`SED_v5/enthalpy_v2.f90::enthalpy_DL01`) returns the total grain enthalpy in erg. The factor $4\pi a^2$ in our form is the grain *surface* area (heating and emission are surface effects), and the $1/C_{\rm tot}$ absorbs the grain volume implicit in the total heat capacity.

2.3 The Monte Carlo loop

Starting from (t_0, T_0), one step of the MC simulation does the following.

1. **Sample the inter-arrival time.** The event-arrival process is Poissonian with rate $\Gamma_{\rm evt}$, so the waiting time $\Delta t = -\ln U_1/\Gamma_{\rm evt}$ where $U_1 \in (0,1]$ is a uniform random number.

2. **Integrate Eq. (7) over $[t_0, t_0 + \Delta t]$.** The grain cools (or, in the rare regime where $H > \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4$, heats) according to the continuous part of the radiation field. During this step we accumulate the time-weighted contribution to the $P(T)$ histogram and to the integrated emitted energy.

3. **Sample the absorbed photon.** Choose a wavelength λ from the normalized PDF $p(\lambda) \propto dN_{\rm abs}/(dt\, d\lambda)$ on $(0, \lambda_c)$, via inverse-CDF lookup on a pre-tabulated CDF.

4. **Apply the enthalpy jump.** The absorbed photon energy $E_{\rm phot} = hc/\lambda$ goes into internal vibrational energy. Solve

$$E_{\rm grain}(T_{\rm after}) = E_{\rm grain}(T_{\rm before}) + E_{\rm phot} \tag{8}$$

for $T_{\rm after}$ by inverting the tabulated $E_{\rm grain}(T)$.

The loop is repeated for a prescribed total number $N_{\rm evt}$ of events (typical: 10^4–5×10^4 per grain for good $P(T)$ statistics), or until a prescribed total simulated time is reached.

2.4 Integrator for the cooling ODE

For a typical $\sim 100\,\text{Å}$ graphite grain in $U = 1$ starlight, the inter-arrival time is $\Delta t \sim 600\,$s while the radiative cooling time after a photon spike is $\sim$ ms. The cooling ODE (7) is therefore mildly stiff: rapid initial decay followed by a long, near-equilibrium tail. Explicit Euler is prohibitively slow because the step size near equilibrium is set by the restoring rate, which is large.

Substantively the right-hand side of Eq. (7) is dominated, after the photon spike, by the T^4 Stefan–Boltzmann emission term: with $H(a) \ll \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4$ the equation reduces to $dT/dt \simeq -(4\pi a^2/C_{\rm tot})\langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4$. Both $C_{\rm tot}(T)$ (Debye) and $\langle Q_{\rm abs}\rangle_T$ are weakly T-dependent, so the trajectory is not a single exponential – it falls steeply at high T and slowly near $T_{\rm eq}$. Any closed-form fit (exponential, power-law T^{-3} from constant-C Stefan–Boltzmann) is an approximation; we therefore integrate Eq. (7) numerically and, when a continuous $T(t)$ is needed for a figure, dump the actual integrator sub-step samples rather than reconstructing from event boundaries.

We use a linearly-implicit (exponential) step. At the current temperature T_0, we form a one-sided finite difference to estimate the local slope

$$\Lambda(T_0) \;\equiv\; \left.\frac{d(dT/dt)}{dT}\right|_{T_0} \approx \frac{(dT/dt)_{1.01\,T_0} - (dT/dt)_{T_0}}{0.01\,T_0}. \tag{9}$$

When $\Lambda < 0$ the dynamics is locally a stable relaxation toward a local equilibrium temperature $T_{\rm eq,loc} = T_0 - (dT/dt)_{T_0}/\Lambda$, and the linearized solution

$$T(t_0 + dt) \;=\; T_{\rm eq,loc} + \left(T_0 - T_{\rm eq,loc}\right) e^{\Lambda\, dt} \tag{10}$$

is exact. We step by

$$dt = \min\!\left(dt_{\rm left},\, \tfrac{\ln 2}{|\Lambda|},\, f_{\rm cap}\, \tfrac{T_0}{|dT/dt|_{T_0}}\right), \qquad f_{\rm cap} = 0.05, \tag{11}$$

once $|T_0 - T_{\rm eq,loc}|/T_0 < 10^{-3}$ we take the full remaining $dt_{\rm left}$ in a single step, since further evolution is unobservable on the bin scale. The first cap, $\ln 2/|\Lambda|$, halves the distance to local equilibrium per step. The third cap, which limits the *fractional* temperature change to $f_{\rm cap} = 5\%$ per step, is essential: the linearized solution (10) is exact only for a locally *linear* ODE, whereas the true cooling is strongly super-linear ($dT/dt \propto -\langle Q_{\rm abs}\rangle_T T^4$). Over a half-life-sized step the

local linearization lags the faster early cooling, so the grain appears to linger at high T, over-populating the hot tail of $P(T)$ and over-emitting in the mid-IR. This bias is most severe for the violently heated tiniest PAH grains and was the dominant source of an MC mid-IR over-prediction before the $f_{\rm cap}$ cap was added (§5.2). With the cap, the linearization error per step is negligible; the step count per cooling segment is bounded by a hard cap (`MAX_STEPS` = 4000 in the fixed-grid `cool_segment`, 300 in the adaptive engines of §7.).
The $P(T)$ histogram and the emitted-energy bookkeeping sub-sample each step at $N_{\rm sub} = 8$ interior points of the sub-step trajectory, each carrying weight $dt/N_{\rm sub}$, so a step that spans several histogram bins deposits its time weight across all of them. For unstable ($\Lambda \geq 0$) or degenerate cases the fallback is a forward Euler step, also subject to the $f_{\rm cap}$ cap.

3. Implementation

3.1 Module layout

The implementation lives under `SEDust/mc/` and is split into small modules that each do one thing. Throughout, cgs units are used (erg, cm, s) except that wavelength is in μm to match the rest of the astrodust pipeline.

- `mc_heatcap.f90`: enthalpy $E_{\rm grain}(T, a)$ and heat capacity $C_{\rm tot}(T) = dE_{\rm grain}/dT$ for selectable compositions (Table 1). $C_{\rm tot}$ is computed by central differences on $E_{\rm grain}$, which works for all backends without composition-specific analytic derivatives.
- `mc_rng.f90`: thread-local pseudo-random number generator (xorshift64*, period $2^{64} - 1$) wrapped in a derived type `rng_t`. Each OpenMP thread owns its own state, so no locking is needed inside the MC inner loop.
- `mc_grain_type.f90`: derived type `mc_grain_t` holding all single-grain pre-tabulated state (λ grid, $Q_{\rm abs}$, u_λ, $H_{\rm cont}$, $\Gamma_{\rm evt}$, photon-wavelength CDF, T grid, $E_{\rm grain}(T)$, $\langle Q_{\rm abs}\rangle_T$). Instances are local to a thread.
- `mc_engine.f90`: the engine. Two setup entry points: `grain_setup_graphite` (D16 graphite sphere Q, used by the standalone validation path) and `grain_setup_from_cabs` (arbitrary $C_{\rm abs}(\lambda)$, used by the SED builder). Three MC drivers with the same signature shape, differing only in the histogram strategy: `mc_run_engine` (the original fixed-grid log $[2, 5000]$ K histogram), `mc_run_engine_2pass`, and `mc_run_engine_buffered`. All three return the time-weighted $P(T)$ histogram (as bin edges $T_{\rm edge}$, dP/dT, and $dP/d\ln T$) and the integrated absorbed and emitted energies; the adaptive engines additionally return the auto-selected grid type (`is_log_out`) and the observed $T_{\rm min}/T_{\rm max}$. Common helpers — `step_advance`, `sub_sample_step`, `build_adaptive_grid`, `bin_sub_samples` — factor out the trajectory math and the log/linear binning so each driver is short and the integrator code is shared. See Section 7. for the algorithm and rationale.
- `mc_sed.f90`: SED builder. Wraps an OpenMP parallel-do loop over the size distribution; each iteration creates its own `mc_grain_t` and `rng_t`, runs MC, and accumulates the thread-local contribution to λI_λ.
- `main_mc.f90`: standalone single-grain driver (`main_mc.x`). Reads a namelist, runs one MC, writes `<prefix>_PT.dat`.

- `main_mc_sed.f90`: size-distribution SED driver (`main_mc_sed.x`). Reads a namelist, calls `sed_init` from the astrodust pipeline to load the grain Q table, size distribution, and PAH cross sections, then loops via `mc_sed`. It calls `sed_init` without the optional `lam_min`, so the wavelength grid is the plain T-matrix Q-table grid (0.0912 µm to 3.98×10^4 µm) and the extreme-ultraviolet extension available to the library does not enter here.

Table 1: Selectable heat-capacity backends (`comp` key).

key	enthalpy model
`gra_dl01`	DL01 `Car0` graphite (Debye-2 with $\Theta_1 = 863$ K, $\Theta_2 = 2504$ K, plus C–H modes for $n_{\rm atom} \le 5.75 \times 10^4$, i.e. $a \lesssim 50$ Å — the Draine & Li 2001 threshold). **Default for graphite and PAH.**
`gra_d85`	Single Debye-3, $\Theta_D = 420$ K, $n_{\rm atom} = 1.14 \times 10^{23}$ cm^{-3}. Original Draine & Anderson (1985) prescription.
`sil_dl01`	DL01 silicate (`Sil`).
`ad_s1_c1`	Astrodust Stage 1, C1 prefactor (pure DL01 `Sil`).
`ad_s1_c2`	Astrodust Stage 1, C2 prefactor (density-corrected).
`ad_s2`	Astrodust Stage 2: $E_{\rm Sil}(N_{\rm Sil}) + E_{\rm Car}(N_{\rm Car})$ via volume-weighted sub-volumes ($f_C^{\rm vol} = 0.193$, $P = 0.20$).

3.2 Validation: energy conservation

For each MC run we accumulate the absorbed energy $E_{\rm abs} = \sum_{\rm events} hc/\lambda$ and the emitted energy $E_{\rm emit} = \int 4\pi a^2 \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4\, dt$ over the full simulation. In steady state these should match. The standalone `main_mc.x` (adaptive two-pass engine) with $a = 100$Å graphite, $U = 1$, $N_{\rm evt} = 2 \times 10^5$ returns $E_{\rm emit}/E_{\rm abs} = 0.997$ – consistent with 1 within the residual from the finite final state. Larger grains, more events, and higher U all push the ratio closer to unity.

For the size-distribution SED builder (`main_mc_sed.x`) on the HD23 astrodust+PAH size distribution at $\log U = 0.20$ ($U = 1.585$), the default-mode population-summed trajectory ratios are now $E_{\rm emit}/E_{\rm abs} = 1.001$ (astrodust) and 1.000 (PAH). The fractional-temperature step cap (Eq. 11) removed the $\sim 10\%$ deficit (0.90/0.93) that the uncapped linearized integrator produced in earlier versions. Independently of the raw trajectory bookkeeping, the SED builder rescales each stochastically heated grain's emission to its absorbed power (§5.2), so the *emitted SED* conserves energy exactly by construction — as the matrix solver does. The all-MC mode, where the large grains carry a small residual that this rescale removes, is discussed in §6.2.

4. Validation against PIIM Figures 24.5 and 24.6

4.1 Figure 24.5: $T(t)$ trajectories

We run the standalone `main_mc.x` with composition `gra_dl01` for the ten panels of Draine (2011) Figure 24.5: grain radii $\{10, 20, 50, 100, 200\}$Å crossed with $U \in \{1, 10^2\}$. Each run is capped at $t_{\rm max} = 10^5$ s ($\sim$1 day) following PIIM Fig. 24.5. The grain starts at $T_{\rm init} = T_{\rm CMB} = 2.725$ K (the code default, so the script does not override it) and warms toward $T_{\rm eq}$ on the first cooling segment; the panels with $N_{\rm evt} = 0$ (e.g. $a = 10$ Å at $U = 1$, where no UV photon arrives within $t_{\max}$) therefore sit on the cooling tail from $T_{\rm init}$ for the whole simulated window. Total wall time across the 10 runs is ~ 1.3 s.

Figure 1 reproduces the qualitative behavior of PIIM Fig. 24.5: large grains ($a \geq 100$Å) sit close to a steady temperature; smaller grains spike to hundreds of K between sparse photon events. The energy balances in each panel are within $\sim 5\%$ for all panels with $N_{\rm evt} > 100$.

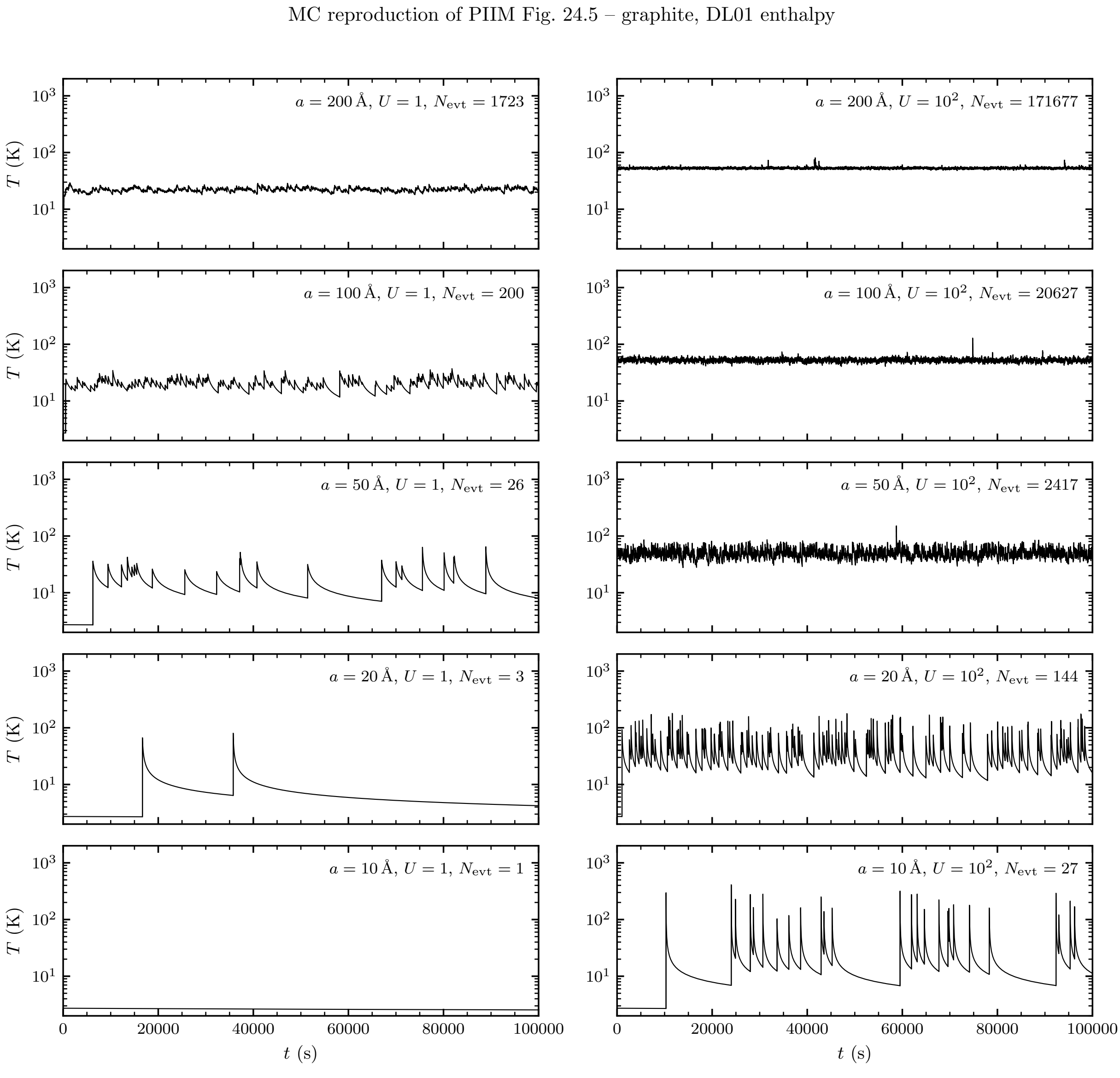

Figure 1: Reproduction of Draine (2011) Figure 24.5: MC temperature histories for graphite grains in the local ISRF. Black curves are the actual cooling trajectories dumped from `cool_segment` at the integrator's adaptive sub-step boundaries (no closed-form reconstruction); vertical line segments show the instantaneous photon-jumps from $T_{\rm pre}$ to $T_{\rm post}$ at each absorption event.

4.2 Figure 24.6: $dP/d\ln T$

For the $P(T)$ reproduction we use the same five sizes at $U = 1$ with $N_{\rm evt} = 2 \times 10^4$ per grain. Figure 2 shows the time-weighted $dP/d\ln T$ distributions, reproducing PIIM Fig. 24.6: large grains have a narrow peak near $T_{\rm eq}$; small grains have broad distributions extending to several hundred K.

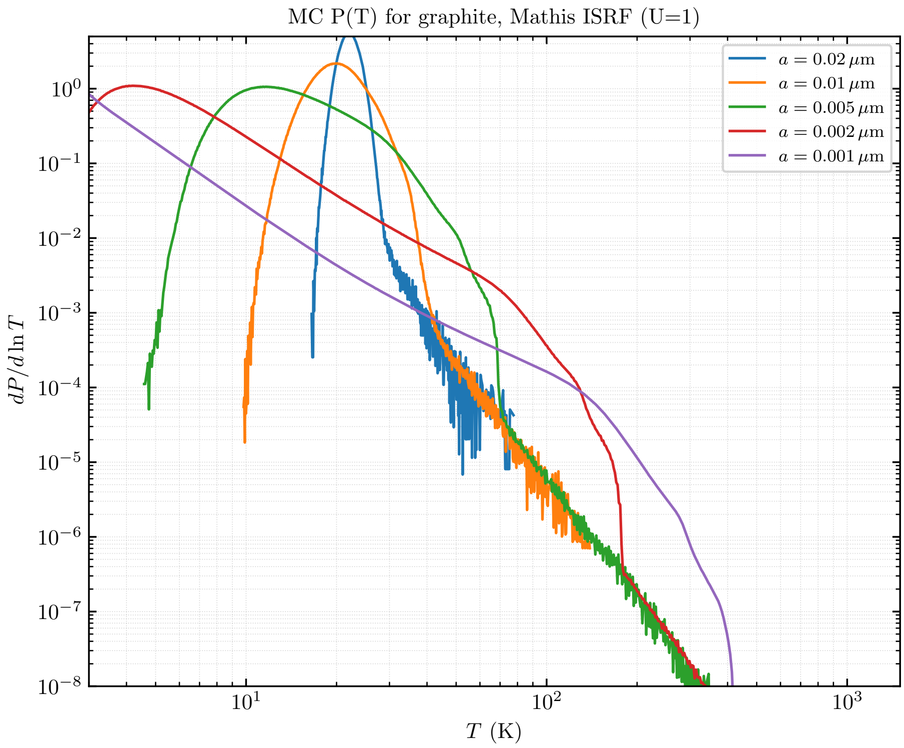

Figure 2: $dP/d\ln T$ for graphite grains in the Mathis ISRF ($U = 1$), for sizes $a \in \{0.001, 0.002, 0.005, 0.01, 0.02\}\,\mu$m. Compare with PIIM Figure 24.6.

5. SED builder over a grain size distribution

The `mc_sed` module integrates $P(T)$ across the HD23 astrodust+PAH size distribution (read by `sed_astrodust_mod::sed_init` from `data/release/size_distribution.dat`) and produces the emission SED $\lambda I_\lambda/N_H$ in the same units as the production `sed_solve` routine.

5.1 SED accumulation

For each grain a_j in the size distribution:

$$J_\lambda \mathrel{+}= (dn/N_H)_j \cdot C_{\rm abs}(\lambda, a_j) \cdot \sum_i \frac{w_i}{t_{\rm total}} \cdot \pi B_\lambda(T_{{\rm mid},i}), \tag{12}$$

where w_i is the time spent in histogram bin i (with center $T_{{\rm mid},i}$) accumulated by the MC. Summing over both grain populations (astrodust + PAH) and applying the same final unit conversion as the production pipeline, $\lambda I_\lambda = \lambda J_\lambda \times 10^{-3}$, gives the final SED in erg s^{-1} cm^{-2} sr^{-1} H^{-1}.

5.2 Energy-conservation rescale and charge-resolved PAH

Two refinements to the raw accumulation (12) are needed to bring the MC SED into band-by-band agreement with the matrix solver. Both were identified by isolating the residual to the PAH component (the astrodust component agreed to $\sim 0.4\%$ from the start) during the cross-check of Section 6..

Single-grain energy-conservation rescale. Equation (12) re-evaluates the Planck function at each bin *midpoint* $T_{\mathrm{mid},i}$. Because $B_\lambda(T)$ is convex in T (the bolometric emission scales as T^4), the midpoint value exceeds the bin-averaged emission — a Jensen-inequality bias that over-weights the hot tail and leaves the grain emitting slightly more than it absorbed. In steady state a grain emits exactly what it absorbs (Kirchhoff), and the matrix solver enforces this by construction. We restore it in the MC builder by rescaling each stochastically heated grain's spectrum by

$$s_{\mathrm{ec}} = \frac{P_{\mathrm{abs}}}{P_{\mathrm{emit}}^{\mathrm{grey}}}, \qquad P_{\mathrm{emit}}^{\mathrm{grey}} = \sum_i w_i 4\pi a^2 \langle Q_{\mathrm{abs}} \rangle_T [T_{\mathrm{mid},i}] \sigma_{\mathrm{SB}} T_{\mathrm{mid},i}^4, \qquad P_{\mathrm{abs}} = \frac{E_{\mathrm{abs}}^{\mathrm{evt}}}{t_{\mathrm{total}}} + 4\pi a^2 H_{\mathrm{cont}}, \tag{13}$$

where P_{abs} is the grain's total absorbed power (discrete photon events plus the continuous long-wavelength heating H_{cont}) and $P_{\mathrm{emit}}^{\mathrm{grey}}$ is the grey bolometric emission of the same histogram. The rescaled single-grain spectrum $s_{\mathrm{ec}} J_\lambda^{(j)}$ then conserves energy exactly while keeping the spectral *shape* set by $P(T)$. Combined with the integrator step cap (§2., Eq. 11), this brought the integrated PAH power from $1.15\times$ the matrix-solver reference (raw midpoint histogram, half-life step) to $1.005\times$.

Charge-resolved PAH populations. The production pipeline treats PAHs as a blend of two charge states, neutral and cation, mixed by an ionization fraction $f_{\mathrm{ion}}(a)$; the two have markedly different cross sections in the near-IR (the cation carries a Mattioda et al. (2005) NIR continuum that the neutral lacks). The matrix solver loops over both charge states — each with its own C_{abs} and size-distribution weight dn — and sums the emission. The MC builder must do the same: `mc_sed_solve_pah` runs the engine twice per size, once with the neutral ($C_{\mathrm{abs}}^{\mathrm{neu}}, dn^{\mathrm{neu}}$) and once with the cation ($C_{\mathrm{abs}}^{\mathrm{ion}}, dn^{\mathrm{ion}}$) at an offset RNG seed, and sums. Using the single charge-averaged cross section instead — as an earlier version did — biases the PAH near-IR (1–5 μm) low by $\sim 23\%$. Resolving the charge states moves the PAH NIR band ratio from 0.77 to 0.97 relative to the reference.

5.3 OpenMP parallelism

The grain loop in `mc_sed::mc_sed_loop` carries an `!$omp parallel do schedule(dynamic,1)` directive. Each thread holds its own `mc_grain_t` and `rng_t` in private storage and contributes to a thread-local `Jout_local(:)`; thread-local totals are merged into the shared accumulator inside a single `!$omp critical` block at the end of the parallel region.

On a 12-core M-series Mac with $N_{\mathrm{evt}} = 5 \times 10^4$ per grain, the population sum over ~ 168 grains (84 sizes $\times$ 2 populations) takes ~ 14 s wall time, about $11\times$ the single-thread cost – the slight loss versus the $12\times$ ideal is the cost of the dynamic schedule with variable work per grain.

6. Comparison with the Guhathakurta–Draine matrix solver

We compare `main_mc_sed.x` against the production astrodust pipeline `calc_sed.x astrodust`. Both implementations read the same Q table (`q_astrodust_P0.20_Fe0.00_1.400.dat`), the same size distribution (HD23 `size_distribution.dat`), the same Mathis ISRF at $\log U = 0.20$, and the same Stage 2 astrodust enthalpy.

6.1 Two solver modes

The `main_mc_sed.x` driver exposes a runtime namelist flag, `use_mc_for_large_grain`. Small (stochastic) grains *always* use the Monte Carlo engine; the flag selects how large (equilibrium) grains are handled:

- `use_mc_for_large_grain = .false.` (default). Each grain is first tested with the equilibrium gate $E_{\rm grain}(T_{\rm eq}) \geq$ 150 eV (same threshold as `sed_solve`). Grains passing the gate are emitted as a delta-function at $T_{\rm eq}$ via `calc_Teq`. Grains failing the gate are simulated with MC.
- `use_mc_for_large_grain = .true.` All grains use MC, regardless of size. This exercises the MC engine on grains where the analytic solver is exact and exposes the integrator bias discussed below.

Figure 3 shows both modes against the reference. In the default mode, MC handles 78 stochastic grains and `calc_Teq` handles 89 equilibrium grains (out of 167 total across the two populations); the resulting SED matches the production `sed_solve` output to within 1–3% in every band. In the `use_mc_for_large_grain=T` mode, all 167 grains go through MC and — with the integrator and SED-builder fixes described below — the SED is essentially identical to the default mode (band ratios agree to $< 0.5\%$ between the two modes). This was not the case in earlier versions, where the all-MC mode underpredicted the FIR peak by $\sim 30\%$; the next subsection traces that bias and the three changes that removed it.

6.2 Energy conservation: the integrator and SED-builder corrections

The $\sim 30\%$ all-MC FIR deficit of earlier versions came from three distinct biases, two opposite in sign, which a naive MC SED builder incurs and which the production matrix solver does not. Tracing them required a component-by-component, band-by-band comparison (the astrodust component agreed from the start; the residual was localized to PAH and to the large equilibrium grains).

1. **Large-grain trajectory collapse (under-emission).** With the static cutoff $\lambda_c = 1000\,\mu$m, every photon a large grain absorbs is treated as a discrete event; but a large grain's heat capacity is so high that even a UV photon raises T by $\ll T_{\rm eq}$, so between events the trajectory relaxes toward the equilibrium of the *continuous* heating alone (essentially $T_{\rm CMB}$) rather than toward $T_{\rm eq}$. The grain then radiates far too little. This is the dominant part of the FIR deficit and is removed by the adaptive cutoff $\lambda_c(a)$ of §7.1, which folds the weak photons into the continuous term so the trajectory equilibrates at the correct $T_{\rm eq}$.
2. **Hot-tail over-population (over-emission).** The linearized exponential step (§2.) is exact only for a locally linear ODE; over a half-life-sized step it lags the super-linear T^4 cooling and the grain lingers hot, over-populating the high-T tail of $P(T)$. For the violently heated tiniest PAHs this over-predicts the mid-IR by up to $\sim 20\%$. The fractional step cap $f_{\rm cap} = 0.05$ (Eq. 11) removes it; with the cap the raw trajectory energy ratio is $E_{\rm emit}/E_{\rm abs} = 1.001$

(astrodust) and 1.000 (PAH) in the default mode, compared with 0.90/0.93 for the uncapped integrator. (In the all-MC mode the raw ratio is 0.97/1.00, the residual $\sim 3\%$ astrodust deficit coming from the large grains whose narrow trajectory the MC still slightly under-resolves; this residual is removed per grain by the rescale below.)

3. **Planck-midpoint convexity (residual over-emission).** Re-evaluating B_λ at bin midpoints over-weights the hot tail by Jensen's inequality. The single-grain energy-conservation rescale of §5.2 (Eq. 13) forces the grey bolometric emission to equal the absorbed power, exactly as the matrix solver does by construction, so the SED conserves energy regardless of the residual.

With all three corrections (plus the charge-resolved PAH of §5.2, which fixes the near-IR), the all-MC mode reproduces the analytic-large-grain reference to the same 1–3% as the default mode (Figure 3). The earlier characterization of the all-MC bias as "an inherent bias of the linearized integrator" was correct for the uncapped step but is no longer the operating regime.

6.3 When to use each mode

Default (`F`). Use for production runs. This is the intended workflow: the MC engine handles the stochastically heated small grains (where it is genuinely needed) and the cheap analytic T_{eq} solver handles the equilibrium grains (where it is exact). Total wall time on 12 cores is comparable to `calc_sed.x astrodust` (single-threaded) and the result reproduces the production SED.

All-MC (`T`). Now a valid cross-check rather than a diagnostic-only mode: with the corrections of §6.2 it reproduces the reference to the same 1–3% as the default mode. The default mode is still preferred for production only because it is cheaper — the analytic T_{eq} solver replaces an MC trajectory for each of the ~ 90 equilibrium grains at no loss of accuracy (those grains are genuinely single-temperature). Use all-MC when an end-to-end MC result is wanted, e.g. to validate the equilibrium gate itself.

6.4 Wall-time scaling

On the 12-core M-series Mac with $N_{evt} = 5 \times 10^4$ per grain, the default mode takes ~ 17 s wall (78 MC grains + 89 analytic). The all-MC mode takes ~ 19 s (all 167 grains MC). By comparison, `calc_sed.x astrodust` (single-threaded) takes ~ 10 s. At $N_{evt} = 10^4$ (still 1–2% statistical precision at most λ) the MC wall times drop to ~ 4–5 s. These figures predate the fractional step cap (Eq. 11); bounding the step to 5% in T raises the sub-step count per cooling segment, which increases the small-grain MC cost by a model-dependent factor of a few. The adaptive engines that produce these SEDs keep `MAX_STEPS` = 300 (the fixed-grid `cool_segment` uses 4000); with the 5% cap the measured truncation rate is negligible (about 6 of 9.4×10^5 cooling segments, 0.0006%, over a full size-distribution run), so 300 remains safe. The MC is therefore now noticeably more expensive per stochastic grain, which is a further reason to keep the cheap analytic T_{eq} branch for equilibrium grains in production.

6.5 Single-grain $P(T)$ cross-check

As a direct single-grain check of the MC temperature distribution against the matrix solver — independent of the size-distribution integral — we run a dedicated driver `main_pT_compare.x` that, for a hand-picked set of grain sizes, calls all three MC engines (`mc_run_engine`, `mc_run_engine_2pass`, `mc_run_engine_buffered`) *and* `calc_P` on the same input ($C_{\rm abs}$, J_λ, enthalpy, kappCMB, T grid) and writes four $dP/d\ln T$ blocks side by side. Figures 4 and 5 below show the adaptive two-pass block; the fixed-grid block agrees with it at the stochastic sizes shown here (its degeneracy issues appear only at $a \gtrsim 80\,\text{Å}$ and are documented in Figure 6). We use $N_{\rm evt} = 5 \times 10^5$ events per grain, which gives smooth MC curves at the 10^{-5} level of $dP/d\ln T$.

Figures 4 and 5 show the comparison for four stochastic-regime sizes of each population. With the step-capped integrator (§2.) the MC and GD distributions overlap to plot accuracy across the body *and* the high-T tail, over the full range of grain sizes — the slight tail suppression seen in earlier (uncapped) versions is gone. The only visible difference is at the extreme cold end, where the adaptive MC grid begins at the lowest temperature the trajectory actually reached while `calc_P` extends to the bottom of the fixed T grid; that region carries negligible weight. Near the peak of each distribution (which dominates the SED) MC and GD agree to within a few percent.

For equilibrium grains ($a \gtrsim 80\,\text{Å}$ in the astrodust population), the wide `T_first` grid (1–3000 K, 250 log-spaced bins) is too broad for `calc_P` to resolve the narrow physical $P(T)$ without the iterative narrowing logic used inside `sed_solve` (`narrow_iterative` in `sed_astrodust.f90`). Calling `calc_P` directly returns $P \equiv 0$ for these grains. This is why our equilibrium-gate branch in the default mode uses `calc_Teq` (delta function at $T_{\rm eq}$) rather than `calc_P` for them. The adaptive two-pass MC solver *can* resolve the narrow $P(T)$ directly because it builds the histogram grid from the observed $(T_{\rm min}^{\rm obs}, T_{\rm max}^{\rm obs})$ of the trajectory itself rather than from a pre-set wide range; see Section 7. for the algorithm and Figure 6 for the resulting $\sim 50\times$ resolution gain over the fixed-grid baseline.

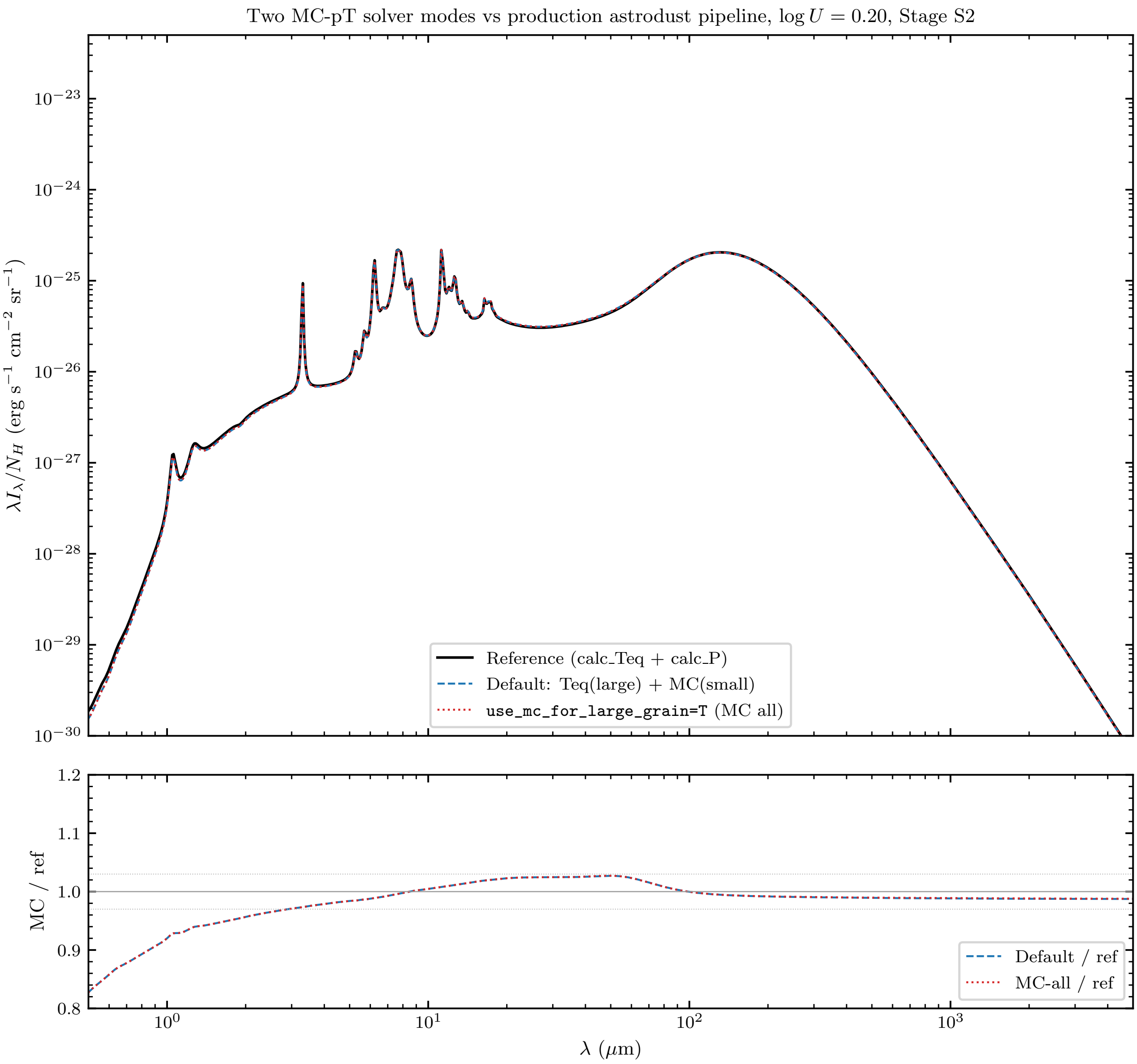

Figure 3: Two MC-pT solver modes versus the production astrodust pipeline, for the HD23 astrodust+PAH model in the Mathis ISRF at $\log U = 0.20$, Stage 2 enthalpy. Both modes use the adaptive 2pass histogram engine, the integrator step cap (§2.), the single-grain energy-conservation rescale and charge-resolved PAH (§5.2), and the adaptive Draine & Anderson (1985) cutoff $\lambda_c(a) = hc/(\delta T_{\rm thresh}\, T_{\rm eq}\, C(T_{\rm eq}))$ with $\delta T_{\rm thresh} = 0.01$ (§7.1). Top: $\lambda I_\lambda/N_H$. Bottom: ratio of each mode to the reference (note the expanded 0.8–1.2 vertical scale). The default mode (blue dashed) and the all-MC mode (red dotted) are indistinguishable and both match the reference to within 1–3% in every band (NIR 0.97, MIR 1.01, FIR 1.00, sub-mm 0.99); the dip below 1 μm is at the optical/far-NIR edge, ~ 5 decades below the peak and carrying negligible power. The $\sim 30\%$ all-MC FIR underprediction of earlier versions — from large-grain trajectory collapse under a static $\lambda_c = 1000\,\mu$m and from hot-tail over-population by the uncapped integrator — is fully removed.

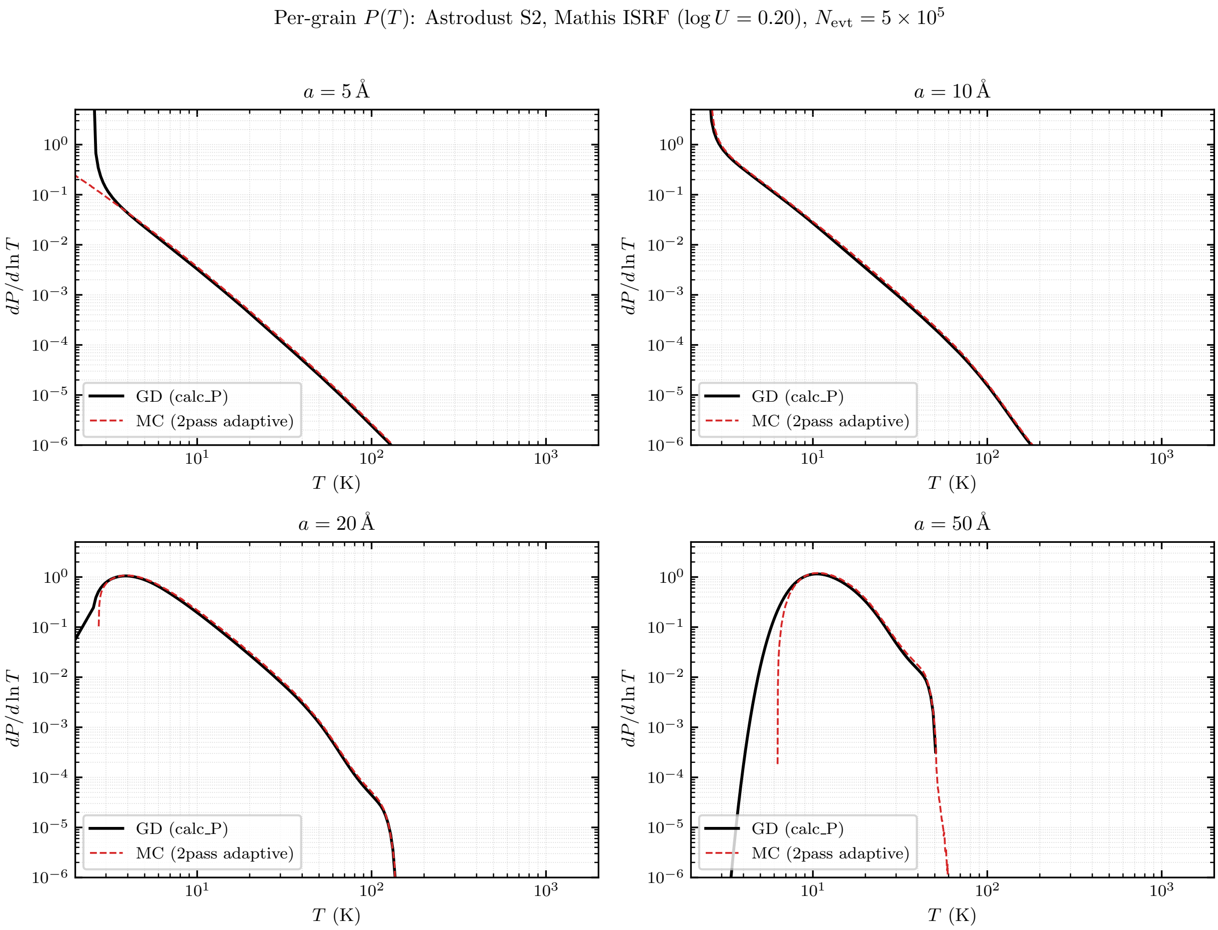

Figure 4: Single-grain $dP/d\ln T$ for the astrodust population (Stage 2 enthalpy), four selected radii in the stochastic regime ($a = 5, 10, 20, 50$ Å). Black: Guhathakurta–Draine matrix solver `calc_P`. Red dashed: Monte Carlo (`mc_run_engine_2pass`, adaptive histogram grid, $N_{\rm evt} = 5 \times 10^5$). The two solvers overlap on the body and the high-T tail of each distribution; they differ only at the cold end, where the adaptive MC grid starts at the trajectory's observed minimum temperature.

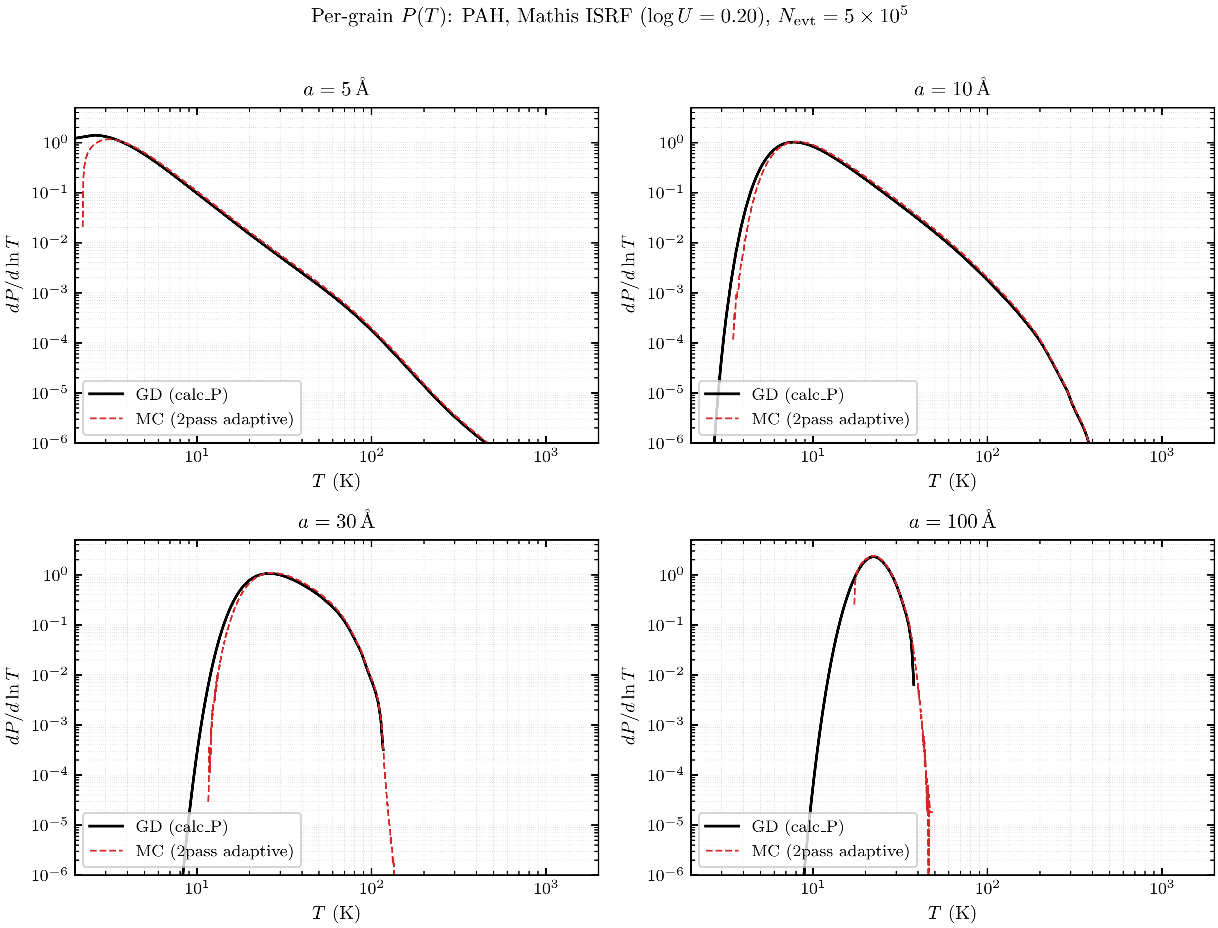

Figure 5: Single-grain $dP/d\ln T$ for the PAH population, four selected radii ($a = 5, 10, 30, 100$ Å). Same conventions as Fig. 4 (adaptive two-pass MC engine, $N_{\rm evt} = 5 \times 10^5$). At $a = 5$ Å the distribution extends to ~ 400 K, where MC and GD agree across five decades of $dP/d\ln T$.

7. Adaptive-grid MC engines

The original engine `mc_run_engine` uses a fixed log histogram on $[T_{\rm min}, T_{\rm max}] = [2, 5000]$ K with $N_{\rm hist} = 600$ bins. This is appropriate for the small (stochastic) grains for which the algorithm was designed: the trajectory spans most of the range and the log spacing matches the multi-decade nature of $P(T)$. For large grains, however, the trajectory stays within a narrow $(T_{\rm eq} \pm \mathcal{O}(\sigma_T))$ band of width $\Delta T/T \sim 1/\sqrt{E_{\rm grain}(T_{\rm eq})}$. At $a = 500$ Å in the local ISRF the entire physical $P(T)$ fits inside $[16.6, 19.5]$ K, which is one bin of the fixed log grid. The histogram becomes a single populated bin at $T_{\rm mid} \approx T_{\rm eq}$, which is fine for the SED integral (Eq. 12; the bin weight times $B_\lambda(T_{\rm mid})$ recovers the correct emission to better than 0.5%) but useless as a diagnostic of the actual distribution shape.
To address this we add two adaptive-grid engines that build the histogram grid from the observed trajectory itself.

Method A: two-pass with deterministic RNG replay (`mc_run_engine_2pass`). Pass 1 runs the MC trajectory exactly as in the original engine but *does not bin*; it only tracks the running $(T^{\rm obs}_{\rm min}, T^{\rm obs}_{\rm max})$ across all sub-step samples after a burn-in of $N_{\rm burn} = \max(50, N_{\rm pass1}/20)$ events (to exclude the initial $T_{\rm init} \to T_{\rm eq}$ transient). At the start of Pass 1 the RNG state is saved (`rng_saved = rng%state`); at the end of Pass 1 the adaptive grid is built from the observed range (with a $\pm5\%$ margin) and the RNG state is restored. Pass 2 re-runs the trajectory for the full $N_{\rm evt}$ events. Because the integrator and the photon-sampling routine are deterministic in their inputs and the RNG is restored to bit-for-bit the same state, Pass 2 traces the identical path as Pass 1 and the only new work is binning. The cost is therefore $\sim 2\times$ the fixed-grid engine. We expose `N_pass1` as an explicit argument but *default to N_pass1 = N_events* (the full trajectory) because shortening Pass 1 to a sub-trajectory makes the adaptive grid built from $N_{\rm pass1} < N_{\rm evt}$ samples too narrow: the remaining Pass 2 samples that fall outside the grid are silently dropped by `bin_sub_samples` and the resulting histogram weight sum drops below unity, biasing the SED low at the FIR/sub-mm (the dropped tail samples are mostly at high T, whose Rayleigh–Jeans-side flux at long λ is linear in T). Empirically, $N_{\rm pass1} = N_{\rm evt}/2$ produced an extra $\sim$ 2–10 percentage-point FIR/sub-mm deficit on top of the integrator-bias deficit; reverting to the full Pass 1 closes that gap to better than 1% across the entire 0.1–3000 μm grid.

Method B: single-pass with sample buffer (`mc_run_engine_buffered`). The trajectory is integrated once. Sub-step pairs $(T_{\rm sub}, dt_{\rm sub})$ after the burn-in are appended to caller-supplied arrays up to a capacity $N_{\rm cap}$ (default 200 000 samples). The running $(T^{\rm obs}_{\rm min}, T^{\rm obs}_{\rm max})$ is tracked online and is *not* bounded by the capacity, so the adaptive grid built at the end uses the true range of the simulated trajectory. Binning runs after the simulation completes. If the buffer fills mid-run (typical for very small grains and very large grains with many sub- steps per cool segment), only the first $N_{\rm cap}$ post-burn-in samples enter the histogram. In steady state this is statistically unbiased; in practice it produces a $\sim$ 1% NIR shift in the SED relative to Method A for the smallest grains. Cost is $\sim$ the same as the original engine.

Adaptive grid spacing. Both methods call `build_adaptive_grid` with margin $\epsilon = 0.05$:

$$T_{\rm lo} = (1-\epsilon)\, T^{\rm obs}_{\rm min}, \quad T_{\rm hi} = (1+\epsilon)\, T^{\rm obs}_{\rm max}. \tag{14}$$

If $T_{\rm hi}/T_{\rm lo} > 3$, the bins are log-spaced from $T_{\rm lo}$ to $T_{\rm hi}$; otherwise they are linearly spaced. The threshold captures the qualitative split between multi-decade stochastic excursions (where log is natural and resolves the high-T tail) and narrow Teq distributions (where linear gives bins of uniform ΔT matched to σ_T). For the case in Fig. 6, the four shown grains span both regimes: $a = 100$ Å picks log (range $[8.4, 35.8]$ K, ratio 4.3); $a = 200$, 501, 1000 Å pick linear (ratios 1.84, 1.17, 1.04). The bin lookup is $O(1)$ in both cases.

Runtime selection. `main_mc_sed.x` exposes a namelist key `mc_engine` $\in$ {`'fixed'`, `'2pass'`, `'buffered'`} that dispatches grain by grain inside the OpenMP loop. The default is `'2pass'` (adaptive). The `'fixed'` engine is unsafe whenever `use_mc_for_large_grain=.true.` because its single populated bin in the equilibrium regime carries no diagnostic content; the driver detects this combination at startup and auto-promotes the engine to `'2pass'` with a logged warning. The SED outputs of the three engines agree to better than 0.5% at every wavelength: the histogram strategy affects $P(T)$ resolution but not the integrated emission, because each bin's contribution is $dP/dT \cdot \Delta T \cdot B_\lambda(T_{\rm mid})$ and the product of weight and Planck function is insensitive to bin refinement once the distribution is reasonably sampled.

Pass 1 vs Pass 2 length in `mc_run_engine_2pass`. The first pass exists only to probe $(T_{\rm min}^{\rm obs}, T_{\rm max}^{\rm obs})$; the routine accepts an explicit `N_pass1` argument but *defaults to N_events* (full trajectory) because shortening Pass 1 to a sub-trajectory makes the adaptive grid built from $N_{\rm pass1} < N_{\rm evt}$ samples too narrow: Pass 2 samples that fall outside the grid are silently dropped by `bin_sub_samples` and the histogram weight sum drops below unity, biasing the SED low at the FIR/sub-mm. Reverting to the full Pass 1 closes that gap to better than 1% across the entire 0.1–3000 μm grid.

7.1 Adaptive DA85 cutoff $\lambda_c(a)$

The Draine & Anderson (1985) stochastic/continuous split (§2.) treats photons with $\lambda > \lambda_c$ as a continuous heating term $H(a)$ and photons with $\lambda < \lambda_c$ as individual stochastic events that produce a temperature jump $\delta T = (hc/\lambda)/C_{\rm tot}(T)$. The split is heuristic: it assumes the individual jumps are large enough to be statistically significant compared to a typical $T_{\rm eq}$ fluctuation. A fixed cutoff $\lambda_c = 1000\,\mu$m, used in earlier versions of this code, is the right scale for small ($a \lesssim 0.1\,\mu$m) astrodust grains but becomes physically wrong for larger grains. As a grows the heat capacity $C_{\rm tot} \propto a^3$ grows fast enough that even UV-photon jumps $\delta T \ll T_{\rm eq}$, and the trajectory between events relaxes toward the local equilibrium of $H_{\rm cont}$ alone (essentially $T_{\rm CMB}$ for Mathis ISRF) rather than toward $T_{\rm eq}$. The result is the trajectory collapse documented in Figure 6's middle–bottom rows (with the static λ_c) and the $\sim 30\%$ FIR underprediction of the all-MC SED.
We replace the static λ_c with an adaptive prescription implemented in `mc_sed_loop` and `main_pT_compare`:

$$\boxed{\lambda_c(a) \;=\; \frac{hc}{\delta T_{\rm thresh}\, T_{\rm eq}(a)\, C_{\rm tot}\big(T_{\rm eq}(a),\, a\big)},} \tag{15}$$

with $\delta T_{\rm thresh} = 0.01$. Photons with λ short enough that their single-event jump exceeds $\delta T_{\rm thresh}\, T_{\rm eq}$ are kept stochastic; everything longer is folded into $H_{\rm cont}$. $T_{\rm eq}(a)$ is supplied by `calc_Teq` on the same single-grain cross sections; $C_{\rm tot}$ is the grain heat capacity at $T_{\rm eq}$. λ_c is clamped to the wavelength grid so the split integrals have at least one bin on each side. For a single a the formula scales as $\lambda_c(a) \propto 1/C_{\rm tot} \propto a^{-3}$, recovering the user's ansatz $\lambda_c(a) = \lambda_{c,0} \cdot (a_{\rm ref}/a)^3$ with the prefactor fixed by the physical threshold rather than by hand. For the HD23 astrodust size distribution at $\log U = 0.20$ the result is $\lambda_c \approx 1000\,\mu$m (clamp) for $a \lesssim 100$ Å (no change from the legacy behavior) and $\lambda_c \approx 0.09\,\mu$m (also clamp, at the short-wavelength edge of the grid) for $a \gtrsim 0.05\,\mu$m, meaning that for all equilibrium grains the full continuous spectrum is folded into $H_{\rm cont}$ and `cool_segment` naturally relaxes to the true $T_{\rm eq}$.

The single-grain diagnostic written by `mc_sed_loop` when `use_mc_for_large_grain=.true.` reports the radiation-equivalent temperature $T_{\rm rad}^{\rm MC} = \langle T^4 \rangle_P^{1/4}$ from the MC histogram alongside the analytic $T_{\rm eq}$. With the static λ_c the ratio $T_{\rm rad}^{\rm MC}/T_{\rm eq}$ falls from 1.0 at $a = 0.03\,\mu$m to 0.22 at $a = 5\,\mu$m. With the adaptive λ_c the ratio is 1.00 ± 0.03 across the entire size range; combined with the integrator step cap and the single-grain energy-conservation rescale (§6.2), the all-MC SED then matches the production reference to 1–3% in every band (Figure 3, red dotted curve).

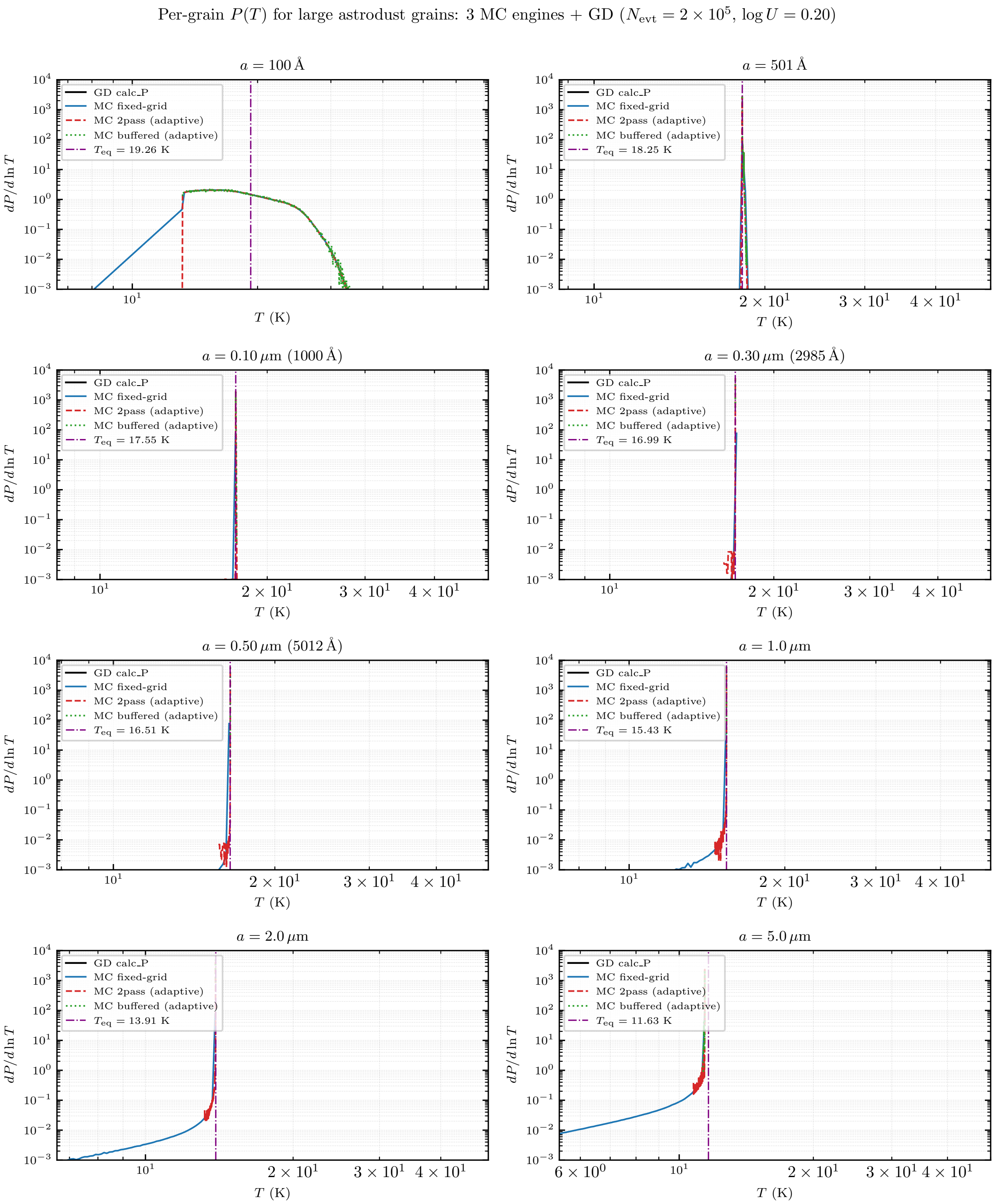

Figure 6: Single-grain $dP/d\ln T$ for large astrodust grains ($a \in \{100, 501, 1000\,\text{Å}, 0.3, 0.5, 1, 2, 5\,\mu\text{m}\}$, Stage 2 enthalpy, Mathis ISRF at $\log U = 0.20$, $N_{\rm evt} = 2\times10^5$). Purple vertical line: $T_{\rm eq}$ from `calc_Teq`. Black: GD `calc_P`; blue: fixed-grid MC; red dashed: 2pass adaptive; green dotted: buffered adaptive. With the adaptive $\lambda_c(a)$ (§7.1) the MC trajectories hold tightly at $T_{\rm eq}$ for every panel from 100 Å to 5 μm; the trajectory collapse toward CMB seen with a static $\lambda_c = 1000\,\mu$m is fully removed.

References

Draine, B. T., & Anderson, N. 1985, ApJ, 292, 494, “Temperature fluctuations and infrared emission from interstellar grains”.

Draine, B. T. 2011, *Physics of the Interstellar and Intergalactic Medium* (Princeton: Princeton University Press), Chap. 24.

Guhathakurta, P., & Draine, B. T. 1989, ApJ, 345, 230, “Temperature fluctuations in interstellar grains. I. Computational method”.

Draine, B. T., & Li, A. 2001, ApJ, 551, 807, “Infrared emission from interstellar dust. I. Stochastic heating of small grains”.

Mattioda, A. L., Allamandola, L. J., & Hudgins, D. M. 2005, ApJ, 629, 1188, “The near-infrared spectrum of polycyclic aromatic hydrocarbon cations”.

Draine, B. T. 2016, ApJ, 831, 109, “Graphite Revisited”.

Hensley, B. S., & Draine, B. T. 2023, ApJ, 948, 55, “The astrodust+PAH model”.